

# 1. Send data to cloud

Sever: <https://thingspeak.mathworks.com>

# 1.1. Create channel

## My Channels

[New Channel](#)

| Name                                                                                                                                                                                      | Created    | Updated          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|
| <b>Parkinson_data_c1</b><br><a href="#">Private</a> <a href="#">Public</a> <a href="#">Settings</a> <a href="#">Sharing</a> <a href="#">API Keys</a> <a href="#">Data Import / Export</a> | 2024-12-30 | 2024-12-30 15:32 |
| <b>Parkinson_data_c2</b><br><a href="#">Private</a> <a href="#">Public</a> <a href="#">Settings</a> <a href="#">Sharing</a> <a href="#">API Keys</a> <a href="#">Data Import / Export</a> | 2024-12-30 | 2025-01-02 08:48 |

## Help

Collect data in a ThingSpeak channel from a device, from another channel, or from the web.

Click **New Channel** to create a new ThingSpeak channel.

Click on the column headers of the table to sort by the entries in that column or click on a tag to show channels with that tag.

Learn to [create channels](#), explore and transform data.

Learn more about [ThingSpeak Channels](#).

## Examples

- [Arduino](#)
- [Arduino MKR1000](#)
- [ESP8266](#)
- [Raspberry Pi](#)
- [Netduino Plus](#)

## Upgrade

Need to send more data faster?

Need to use ThingSpeak for a commercial project?

[Upgrade](#)

## New Channel

**Name**

**Description**

Field 1  ☒

Field 2  ☐

Field 3  ☐

Field 4  ☐

Field 5  ☐

Field 6  ☐

Field 7  ☐

Field 8  ☐

**Metadata**

1. Sign up a free account
2. Create new channel

The number of variables send to channel for free account

# 1.2. APIS KEY

## Example

Channel ID: 2799487  
Author: mwa0000036384845  
Access: Private

API key like a address to write, read or update data to channel

Private ViewPublic ViewChannel SettingsSharingAPI KeysData Import / Export

Write API Key

Key

KG4MCK3FGN6NOC DG

Generate New Write API Key

Read API Keys

Key

AZBRR0AP0V0TLLVL

Note

Save Note

Delete API Key

Add New Read API Key

Help

API keys enable you to write data to a channel or read data from a private channel. API keys are auto-generated when you create a new channel.

API Keys Settings

- Write API Key:** Use this key to write data to a channel. If you feel your key has been compromised, click **Generate New Write API Key**.
- Read API Keys:** Use this key to allow other people to view your private channel feeds and charts. Click **Generate New Read API Key** to generate an additional read key for the channel.
- Note:** Use this field to enter information about channel read keys. For example, add notes to keep track of users with access to your channel.

API Requests

Write a Channel Feed

GET https://api.thingspeak.com/update?api\_key=KG4MCK3FGN6NOC DG&field1=0

Read a Channel Feed

GET https://api.thingspeak.com/channels/2799487/feeds.json?api\_key=AZBRR0AP0V0TLLVL

Read a Channel Field

GET https://api.thingspeak.com/channels/2799487/fields/1.json?api\_key=AZBRR0AP0V0TLLVL

# 1.3. Interface to observe data

## Example

Channel ID: 2799487  
Author: mwa000036384845  
Access: Private

Private View Public View Channel Settings Sharing API Keys Data Import / Export

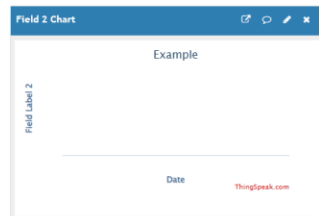
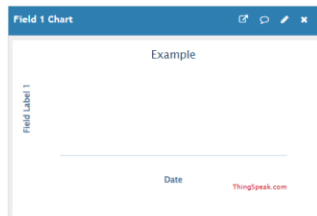
+ Add Visualizations + Add Widgets Export recent data

MATLAB Analysis MATLAB Visualization

Channel 3 of 3 < >

## Channel Stats

Created: 5.minutes.ago  
Entries: 0



The data will be presented on graphs.  
One graph one field (variable)



## Parkinson\_data\_c2

Channel ID: 2797719  
Author: mwa000036384845  
Access: Private

Private View Public View Channel Settings Sharing API Keys Data Import / Export

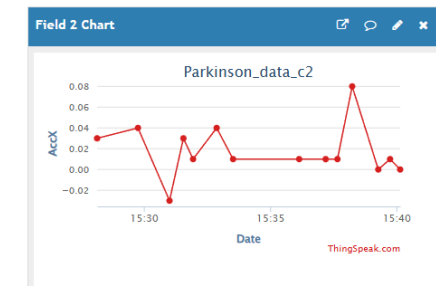
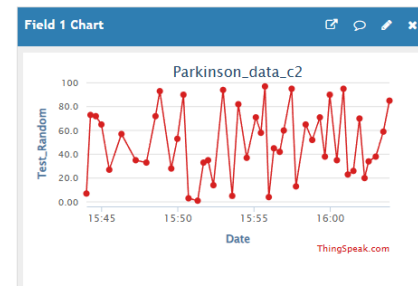
+ Add Visualizations + Add Widgets Export recent data

MATLAB Analysis MATLAB Visualization

Channel 2 of 3 < >

## Channel Stats

Created: 2.days.ago  
Last entry: less.than.a.minute.ago  
Entries: 85



# 1.4. Limitation time collect data

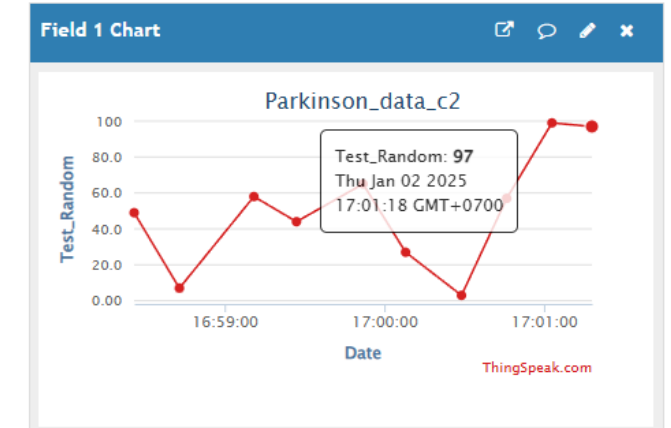
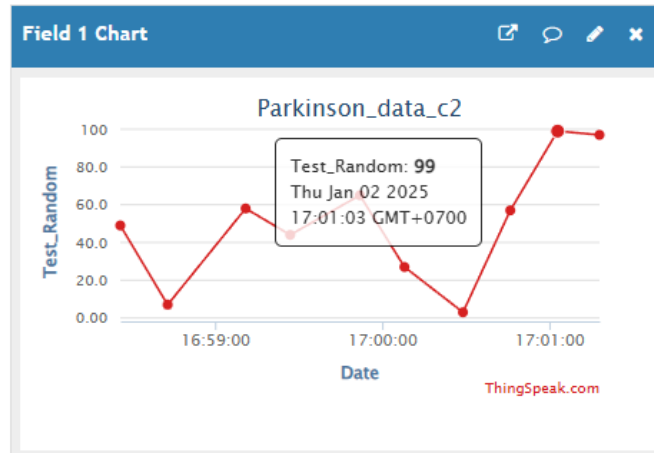
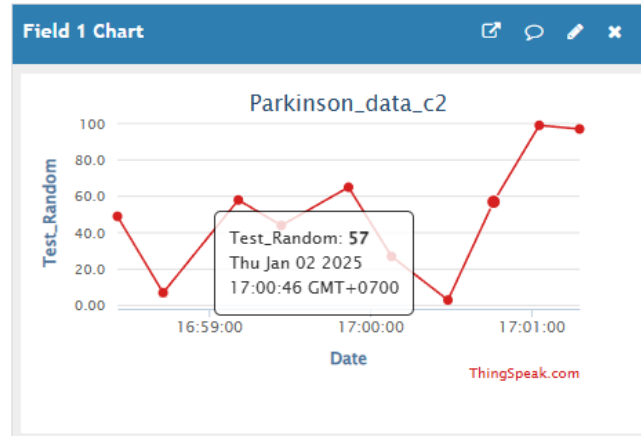
| ThingSpeak™ Channels ▾ Apps ▾ Devices ▾ Support ▾ Commercial Use How to Buy CT |                                               |                                                                                                               |
|--------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|                                                                                | FREE<br>For small non-commercial projects     | ACADEMIC<br>For academic use by faculty, staff, or researchers at degree-granting institutions <sup>(1)</sup> |
| Scalable for larger projects                                                   | ✗<br>No. Annual usage is capped.              | ✓                                                                                                             |
| Number of messages                                                             | 3 million/year<br>(~8,200/day) <sup>(2)</sup> | 33 million/year per unit<br>(~90,000/day per unit) <sup>(2)</sup>                                             |
| Message update interval limit                                                  | Every 15 seconds                              | Every second                                                                                                  |
| Number of channels                                                             | 4                                             | 250 per unit                                                                                                  |
| MATLAB Compute Timeout                                                         | 20 seconds                                    | 60 seconds                                                                                                    |
| Private channel sharing                                                        | Limited to 3 shares                           | Unlimited                                                                                                     |
| Technical Support                                                              | Community Support                             | Standard MathWorks support                                                                                    |
| Max image size                                                                 | ✗<br>Image feature unavailable                | 5 MB                                                                                                          |
| Messages used per image                                                        | ✗                                             | 100                                                                                                           |

<sup>(1)</sup>Not for government, commercial or other organizational use.

<sup>(2)</sup>Based on constant update rate.

Free account only updates data per 15 seconds

# 1.4. Limitation time collect data



With three last samples, average time update data 16s

# 1.4. Limitation time collect data

## Pricing calculator



How many channels?\*

Currently: 3

To be added:



How often will they collect data?

Every  Seconds ▾

Calculated number of channels needed: 3

| Calculated Message Use** |
|--------------------------|
| 94,608,000/year          |
| 7,776,000/month          |
| 259,200/day              |
| 10,800/hour              |
| 180/minute               |

Our Recommendation:

Thingspeak units: 3

License type: Academic

Maximum number of channels allowed: 750

\* Typically one channel per device

\*\* Assumes constant message rate

## Purchase

License type: Academic

ThingSpeak units:

x USD 275.00  
price/unit/year (2-5 units)

Total: USD 825.00/year

[Purchase](#)

You will be taken to the MathWorks store to complete your purchase.

Questions about ThingSpeak purchasing?

[? Licensing FAQ](#) [✉ Contact us](#)

**Conclusion:** This is the limit of receiving data of Thingspeak but still does not meet the condition of measuring the maximum frequency for the patient which is 20Hz.

## 2. Module Wifi ESP8266 UART ESP-01S Ai-Thinker

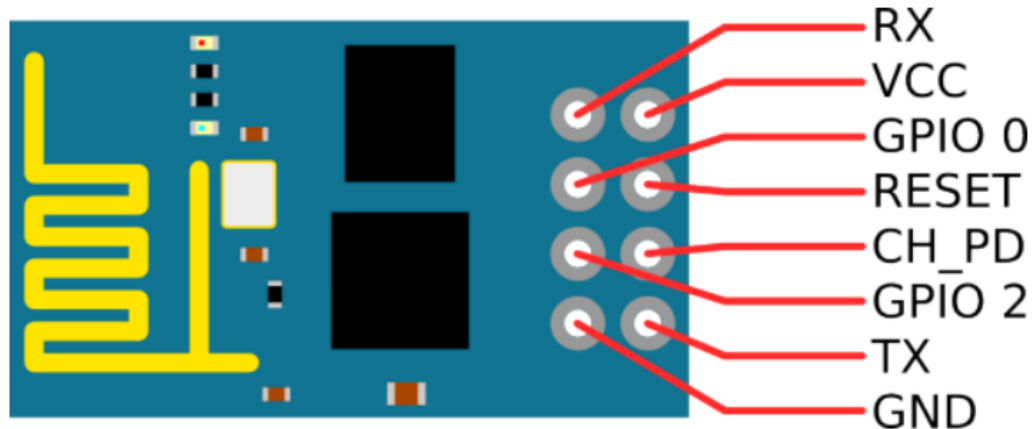
Basically, this module is used to connect to wifi and send data to sever.

Reference: [https://docs.ai-thinker.com/\\_media/esp8266/docs/esp-01\\_product\\_specification\\_en.pdf](https://docs.ai-thinker.com/_media/esp8266/docs/esp-01_product_specification_en.pdf)  
<https://www.microchip.ua/wireless/esp01.pdf>  
<https://hshop.vn/mach-thu-phat-wifi-esp8266-v1>



## 2.1. Hardware

Sơ đồ chân :



1. **VCC: 3.3V** , dòng có thể lên **300mA** vì thế cần mạch nguồn riêng **ams1117 5V->3.3V**
2. GND: 0V
3. Tx: Chân Tx của giao thức UART, kết nối đến chân Rx của vi điều khiển.
4. Rx: Chân Rx của giao thức UART, kết nối đến chân Tx của vi điều khiển.
5. RST: chân reset, kéo xuống mass để reset.
6. **CH\_PD: chân này nếu được kéo lên mức cao module sẽ bắt đầu thu phát wifi, kéo xuống mức thấp module dừng phát wifi. Vì ESP8266 khởi động hút dòng lớn nên chúng ta giữ chân này ở mức 0V khi khởi động hệ thống của mình , sau 2 s hãy kéo chân CH\_PD lên 3.3V, để đảm bảo module hoạt động ổn định.**
7. GPIO0: kéo xuống thấp cho chế độ upgrade firmware.
8. GPIO2: không sử dụng.

## 2.2. Configuration

```
#include <SoftwareSerial.h>
#define RX 8
#define TX 10
String AP = "LAU_3";          // AP NAME
String PASS = "nhatro29almn#L3"; // AP PASSWORD
String API = "ZY4AS3D7QC7IMFEC"; // Write API KEY
String HOST = "api.thingspeak.com";
String PORT = "80";
String field = "field1";
int countTrueCommand;
int countTimeCommand;
boolean found = false;
int valSensor = 1;
```

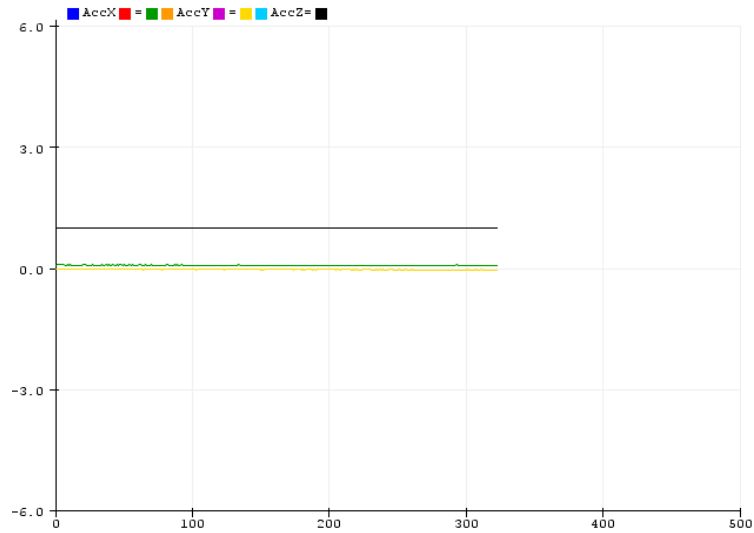
# 3. Collect data from MPU6050

Datasheet:

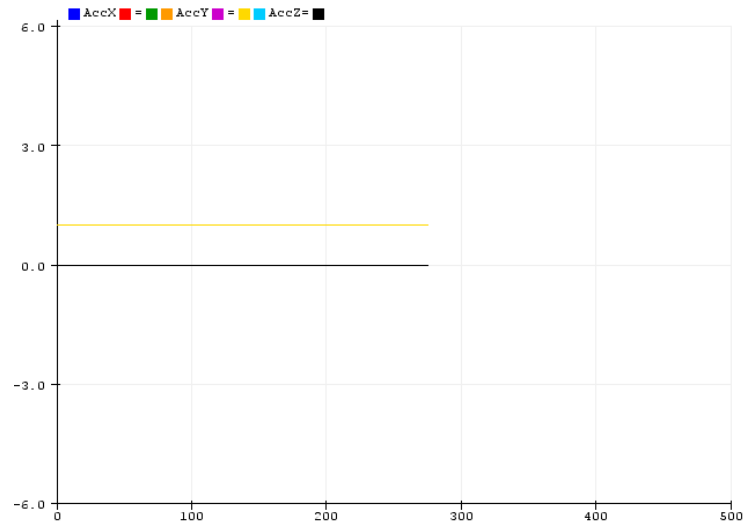
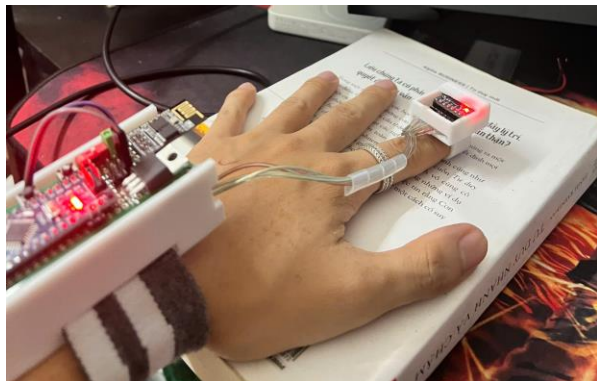
<https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Datasheet1.pdf>

<https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Register-Map1.pdf>

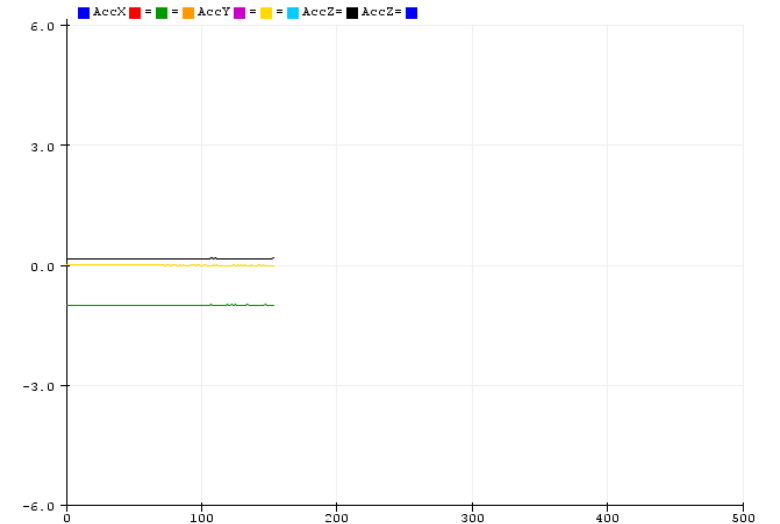
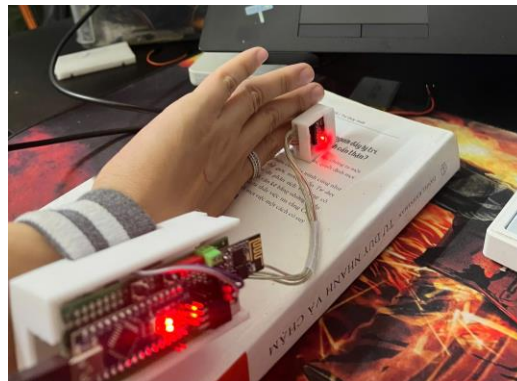
# 3.1. Acceleration



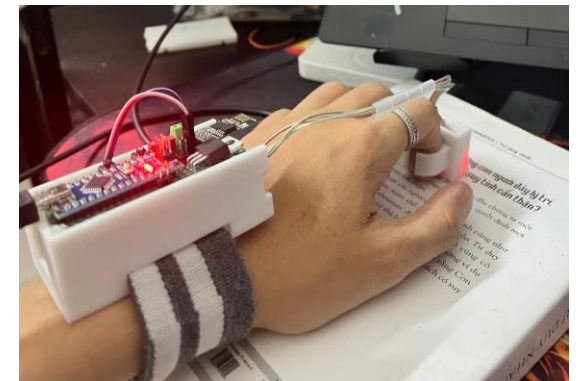
AccZ = 1g



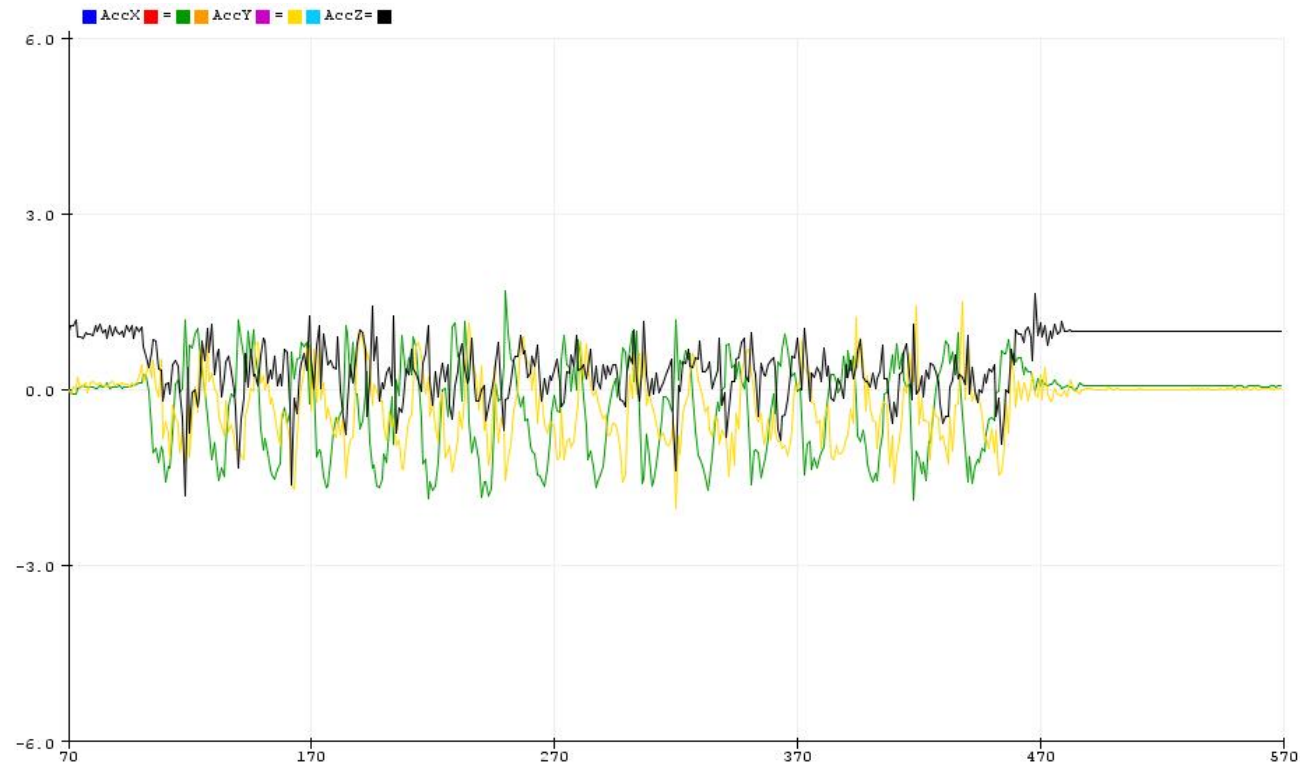
AccY = 1g



Accx = -1g

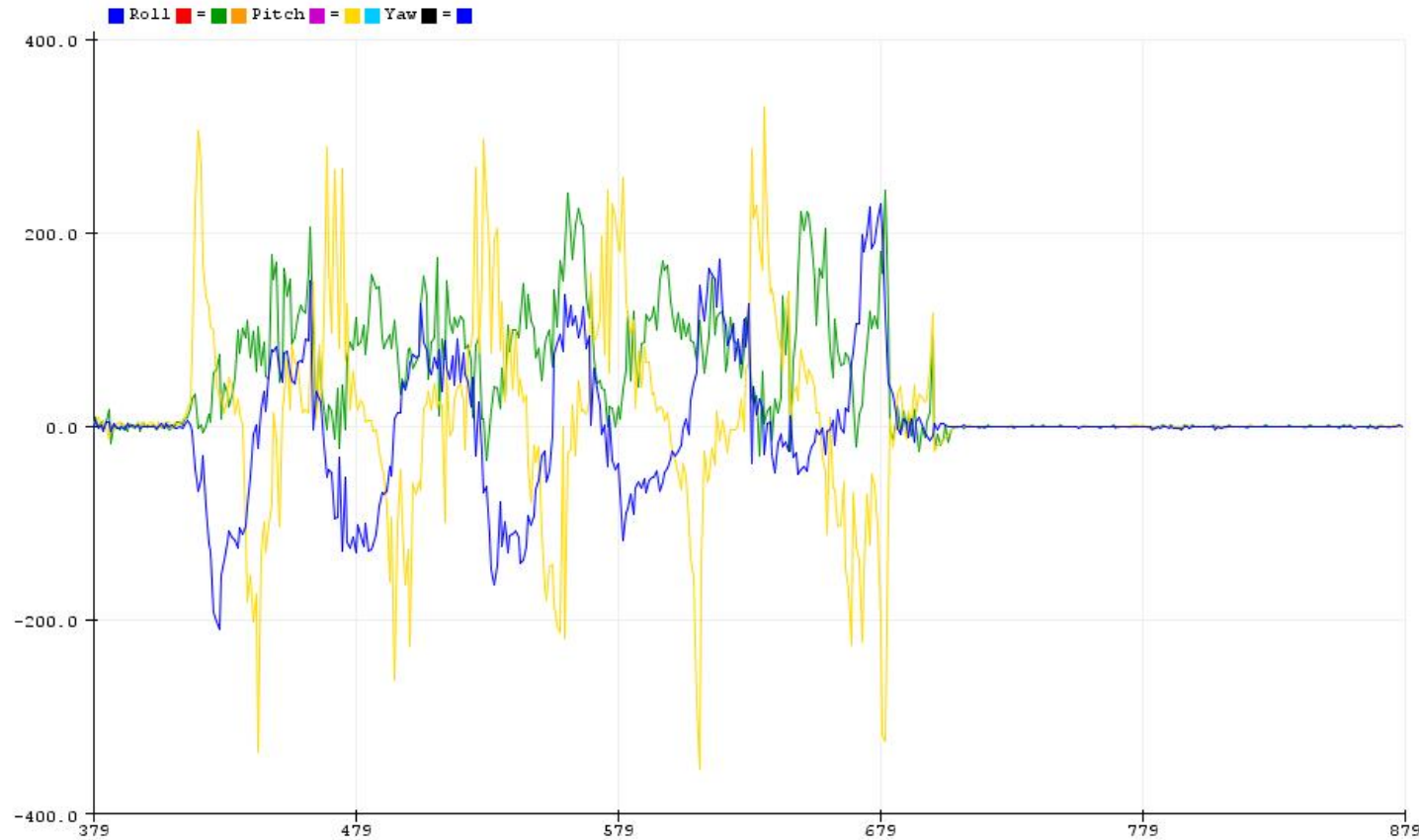


## 3.2. Test sensor



Acceleration (g) of the pronation-supination movements of the right hand (PSMR) (Gia tốc khi quay cổ tay trái)

## 3.2. Test sensor



Angular velocity (degree/s) of the pronation-supination movements of the right hand (PSMR) (Vận tốc góc khi quay cổ tay trái)

## 3.3. Conclusion

- It is feasible to use mpu6050 to read acceleration, velocity and rotation angle.